Activity 3

6/8/26

1. given $Gx = 24$
   $Gy = 18$

1) Gradient magnitude: $\sqrt{Gx^2 + Gy^2}$

$$= \sqrt{(24)^2 + (18)^2}$$

$$= \sqrt{576 + 324}$$

$$= \sqrt{900} = \sqrt{(30)^2} = 30$$

2) Gradient direction: $\theta = \tan^{-1}\left(\frac{Gy}{Gx}\right)$

$$\theta = \tan^{-1}\left(\frac{18}{24}\right)$$

$$\theta = \tan^{-1}(0.75)$$

$$\boxed{\theta = 36.87^\circ}$$

a) given

pixel	Magnitude
p1	85
p2	50
p3	90
p4	60
p5	30

since the gradiant direction is horizontal, compare each pixel with its left and right neighbors.

P1 → Border pixel → supressed (0)

P2 → 50 ⇒ compare with 25 and 90
Not maximum so (0)

P3 → 90 ⇒ compare with 50 and 60
It is maxium so (90)

P4 → 60 ⇒ compare with 90 and 80
Not max so (0)

P5 ⇒ 50 ⇒ border so (0) supressed

Pixel	P1	P2	P3	P4	P5
output	0	0	90	0	0

3. given

High Threshold = 100

Low Threshold = 50

Pixels & Magnitude		classification	
135	P1	P1 = 135 > 100	Strong edge
90	P2	P2 = 90 < 100	weak
45	P3	P3 = 45 < 100	discard
120	P4	P4 = 120 > 100	Strong
70	P5	P5 = 70 < 100	weak
20	P6	P6 = 20 < 100	discard

Strong edges : P1 , P4

weak edges : P2 , P5

NON edges : P3 , P6

4. given

pixel	magnitude
P1	135
P2	95
P3	40
P4	115
P5	65

P6 25

P7 105.

HT = 100

LT = 50

given P2 is connected to P1

P5 is not connected to any strong edges.

strong edges : P1, P4, P7

weak edges : P2, P5

Non edges : P3, P6

Hysteresis :

P2 is connected to strong edge P1

so keep as edge

P5 is not connected to any edge

so Remove

Final edge pixels after Hysteresis

P1, P2, P4, P7